

Phase 5 – Project Development Phase

Project Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Introduction

The Project Development phase focuses on implementing the planned features in the ServiceNow platform. During this phase, users, roles, custom tables, fields, and Access Control Lists (ACLs) were created and configured to enforce secure record-level access based on the user's role and the Branch field.

2. Development Environment

Component	Details
Platform	ServiceNow Developer Instance
Language	JavaScript (ACL Script)
Security	Access Control Lists (ACLs)
Database	ServiceNow Custom Table
Browser	Google Chrome

3. User Creation

A test user named EEE User was created with the following details.

Field	Value
User ID	EEE User
First Name	EEE
Last Name	User
Email	eeeuser@gmail.com

4. Role Creation

Four custom roles were created.

Role	Purpose
bb1	Read Access
bb2	Create Access
bb3	Write/Edit Access
bb4	Delete Access

All four roles were assigned to the EEE User.

5. Custom Table Development

A custom table named Institution Details (u_institution_details) was created.

Table Fields

Field Name	Data Type
Student Roll Number	Auto Number
Student Name	Reference (User)
Faculty Name	Reference (User)
Branch	Choice
Email	String
Phone Number	String
Description	Multi-line String

6. Sample Record Creation

Multiple student records were created using different branch values.

Student	Branch
Student 1	EEE
Student 2	ECE
Student 3	CSE

These records were used for testing ACL behavior.

7. ACL Implementation

Read ACL

Property	Value
Type	Record
Operation	Read
Table	u_institution_details
Role	bb1
Condition	Branch = EEE

Script Used

JavaScript

```
(function () {  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
  
    if (gs.hasRole('bb1')) {  
        return true;  
    }  
  
    return false;  
})();
```

Create ACL

- Role: bb2
- Operation: Create

Write ACL

- Role: bb3
- Operation: Write

Delete ACL

- Role: bb4
- Operation: Delete

8. Development Workflow

The implementation followed these steps:

1. Created the EEE User.
2. Created four custom roles.
3. Assigned roles to the user.
4. Created the Institution Details table.
5. Added all required fields.
6. Inserted sample records.
7. Configured Read ACL with JavaScript.
8. Created Create, Write, and Delete ACLs.
9. Verified permissions using impersonation.

9. Development Challenges

Challenge	Solution
Role assignment	Verified assigned roles
ACL configuration	Tested using impersonation
Record visibility	Checked Branch condition
Permission errors	Reviewed ACL settings

10. Development Outcome

The development phase successfully implemented a secure ServiceNow application where administrators retained full access, users with bb1 could view EEE records, bb2 enabled record creation, bb3 allowed editing, and bb4 permitted deletion. The application achieved role-based record security through Script-Controlled ACLs.